

Numerical Controllability of Parabolic PDEs

Research Direction, Experimental Program, and PhD Roadmap

Central research question

How does a particular numerical control formulation behave under space-time refinement, and can one obtain sharp, practical and reliable bounds on control cost, conditioning, approximation error, and computational work?

Proposed research direction

Reliable numerical controllability of parabolic PDEs: quantitative approximation error, conditioning, solver accuracy, regularization, and computational cost under space-time refinement, with adaptive and preconditioned algorithms.

Why this is potentially novel

- A simple convergence study alone is not sufficient: several existing papers already establish forms of discrete or relaxed controllability and convergence.
- The 1-D heat equation is an ideal benchmark because its continuous controllability theory is mature, making numerical phenomena easier to isolate.
- A stronger numerical-analysis contribution is to quantify the interaction among discretization, optimization, regularization, conditioning, and computational cost.
- The eventual extension to viscous Burgers provides a nonlinear stress test rather than being the starting point.

Target error decomposition

$$\|u - u_{h,k}\| \leq E_{\text{space}} + E_{\text{time}} + E_{\text{opt}} + E_{\text{reg}}.$$

For nonlinear Burgers problems, additional nonlinear discretization and stabilization effects may need to be included.

Core literature context

Area	Representative work	Main relevance
Primal / least-squares heat control	Fernández-Cara & Münch; Münch & Puel (2001)	Addresses difficulties of classical HUM; gives numerical convergence and conditioning estimates.
Penalized HUM	Boyer (2013)	Connects discretization, regularization and relaxed/null controllability.
Fully discrete Carleman	González Casanova & Hernández-Santana (2021)	Relaxed observability and mesh/time restrictions.
Parabolic optimal-control error analysis	McIntyre & Vexler; a posteriori literature	Provides established state/control error-estimation foundations.
Nonlinear heat	Fernández-Cara & Münch; Lemoine, Marin, Bayle & Milot (2019)	Useful for building knowledge toward nonlinear control.
Burgers controllability	Fernández-Cara & Guerrero; Glas & Guerrero (2020)	Provides a natural benchmark and nonlinear stress test.

Four-Month Experimental Plan

Month 1 — Experimental laboratory / reproducibility

- Implement a clean baseline: 1-D heat equation + HUM + implicit Euler.
- Create an experiment-tracking layer so every run is reproducible.
- Store parameters, numerical outputs, solver statistics, operators/matrices where practical, controls, states, plots, code commit, and machine metadata.
- Build a lightweight frontend rather than treating software itself as the research contribution.
- Goal: run roughly 100 experiments reproducibly without manual file/script management.

Month 2 — Fundamental refinement

- Fix a very fine time discretization and refine space: $N = 20, 40, 80, 160, 320, 640$.
- Fix space and refine time: $M = 50, 100, 200, 400, 800, 1600$.
- Test coupled refinement $k = C h^\beta$ with $\beta = 0.5, 1, 1.5, 2$.
- Track condition number, minimum/maximum eigenvalues, control cost, terminal error, control error, iterations, CPU time and memory.
- Goal: determine whether clean and reproducible scaling laws emerge.

Month 3 — Explain the mechanisms

- Sweep regularization, e.g. $\varepsilon = h^p$ for $p = 0, \dots, 8$.
- Sweep optimization tolerances, e.g. 10^{-4} through 10^{-12} .
- Test preconditioning and compare spectral and iteration behavior.
- Vary T , diffusion/viscosity, control-region size/location, and initial-condition frequency.
- Build an error budget separating discretization, optimization, and regularization effects.

Month 4 — Formulation comparison + Burgers pilot

- Implement a least-squares/variational heat-control formulation.
- Run matched experiments against HUM.
- Compare spectra, condition numbers, control error, terminal error, iterations, CPU time and memory.
- Pilot viscous Burgers at low amplitude and moderate viscosity before increasing nonlinearity.
- Goal: determine whether the experimental framework survives nonlinear convection.

First experiment: deliberately narrow

Start with 1-D heat + HUM + implicit Euler. The first objective is not to explore every parameter, but to establish a clean baseline and identify a reproducible numerical phenomenon.

Record: $\kappa(A_h)$, $\lambda_{\min}$, $\lambda_{\max}$, $\|u_h\|$, $\|y_h(T)\|$, CG iterations, CPU time, and control error against a manufactured or high-resolution reference.

Experiments that become important after the baseline

- Manufactured-control experiment with known exact state/control.
- Independent space and time refinement, followed by coupled refinement.
- Solver-tolerance study to determine when optimization error becomes negligible compared with discretization error.
- Condition-number and eigenvalue scaling laws.
- Regularization study.
- Preconditioning: diagonal/Jacobi, incomplete factorizations, AMG or operator-based alternatives.
- HUM versus least-squares/variational formulations.
- High-frequency initial data $y_0 = \sin(m\pi x)$.
- Small-time regime $T \rightarrow 0$.
- Control-region size and location.
- Diffusion/vanishing-viscosity regime.
- Floating-point precision and roundoff limits.
- Cost-versus-accuracy and computational-complexity curves.

Experimental Tracking / W&B-MLflow-Style; Frontend

A lightweight web application can sit above the numerical solver and function as a research instrument.

Architecture: numerical solver → experiment runner → results database → analysis layer → web frontend.

Each run should record

Category	Examples
PDE	heat, Burgers
Geometry / dimension	1-D, domain, mesh size
Control	internal/boundary, control region
Time	T, M, Δt
Space	N, h
Formulation	HUM, least-squares
Regularization	ϵ and scaling rule
Solver	CG/GMRES, tolerance, preconditioner
Outputs	control cost, terminal error, control error
Linear algebra	$\lambda_{\min}$, $\lambda_{\max}$, condition number, iterations
Reproducibility	run ID, git commit, machine metadata

Suggested dashboards

- Mesh refinement: error and control convergence versus h and Δt .
- Conditioning: $\lambda_{\min}$, $\lambda_{\max}$ and κ versus resolution.
- Error decomposition: spatial, temporal, optimization and regularization contributions.
- Cost versus accuracy: control cost and computational work.
- Formulation comparison: HUM versus least-squares.
- Heat versus Burgers: how nonlinear convection changes the numerical picture.
- Run provenance: click any result to recover all parameters and reproducibility metadata.

Potential PhD-level contributions

- Sharp or practically useful scaling laws for conditioning and control cost under refinement.
- A computable error budget for numerical controls that separates discretization, optimization and regularization errors.
- A criterion linking solver tolerance to discretization accuracy, avoiding unnecessary linear/nonlinear iterations.
- Mesh/time refinement rules tailored to controllability computations.
- Preconditioners with provable or experimentally demonstrated mesh-independent/improved iteration behavior.
- A systematic numerical comparison of HUM and primal least-squares formulations.

- Extension of the resulting framework to nonlinear viscous Burgers.

Strategic research storyline

Reproduce → Measure → Quantify → Explain → Precondition → Adapt → Extend to Burgers.

The key distinction is between demonstrating that a numerical control exists and understanding whether the computed control is reliable, accurately resolved, computationally tractable, and stable as the discretization changes.

Possible long-term theorem/algorithm target

Develop a computable estimator of the form $E_{\text{total}} \leq C(\eta_{\text{space}} + \eta_{\text{time}} + \eta_{\text{opt}} + \eta_{\text{reg}})$, together with a preconditioning strategy for which $\kappa(P^{-1}A_{h,k})$ remains bounded or has a provably improved dependence on h and k .

This document is a research-planning synthesis based on the literature and experimental direction discussed in the project.